INCH-POUND

MIL-T-52340B
9 September 1991
SUPERSEDING
MIL-T-52340A(ME)
8 January 1974

MILITARY SPECIFICATION

TOOL, VERTICAL CONNECTING:

ERECTION EQUIPMENT FOR BRIDGE,

FLOATING, PNEUMATIC FLOAT, CLASS 60

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a vertical connecting tool to be used as an erection aid in the construction of the class 60 floating bridge.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir, Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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SPECIFICATIONS

FEDERAL

FF-S-200	-	Set Screws: Hexagon Socket and Spline Socket, Headless.
RR-C-271	-	Chains and Attachments, Welded, Weldless, and Roller Chain.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple Wall.
PPP-T-60	-	Tape: Packaging, Waterproof.

MILITARY

MIL-T-704	-	Treatment and Painting of Materiel.
MIL-H-5606	-	Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.
MIL-H-6083	-	Hydraulic Fluid, Petroleum Base, for Preservation and Testing.
MIL-C-16173	-	Corrosion Preventive Compound, Solvent Cutback, Cold-Application.

STANDARDS

FEDERAL

FED-STD-595	-	Colors.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-889	-	Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

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DRAWINGS

ME

C7940	-	Tool, Vertical Connecting: Erection Equipment for Bridge, Floating, Pneumatic Float, Class 60.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-JBS, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 36/A 36M	-	Structural Steel.
A 572/A 572M	-	High Strength Low-Alloy Columbium-Vanadium Steels of Structural Quality.
D 3953	-	Strapping, Flat Steel and Seals.
D 4675	-	Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

A5.1	-	Covered Carbon Steel Arc Welding Electrodes.
A5.18	-	Carbon Steel Filler Metals for Gas Shielded Arc Welding.
D1.1	-	Structural Welding Code - Steel.

(Application for copies should be addressed to the American Welding Society, 550 North West LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION (NMFTA)

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 2200 Mill Rd., Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The vertical connecting tool shall be as shown on drawing C7940-1 and as specified herein.

3.1.1 Drawings. The drawing forming a part of this specification is an end product drawing. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and as shown on the drawing. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3.1 Materiel deterioration prevention and control. The item(s) shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the item may be exposed.

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3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Structural steel. Structural steel shall conform to ASTM A 36/A 36M.

3.3.5 Low-alloy structural steel. Low-alloy structural steel shall be conform to ASTM A 572/A 572M, grade 50.

3.4 Hydraulic ram. The hydraulic ram shall have a capacity of 7 tons with an internal pressure between 5900 and 6700 psi. The materials used in the hydraulic ram shall be compatible with fluid conforming to MIL-H-6083 or MIL-H-5606. When shipped, the hydraulic ram shall not be filled with hydraulic fluid except as specified in 5.1.1 and 5.1.2.

3.5 Hydraulic couplings. A set of hydraulic couplings shall consist of a female section for attachment to the ram: male section for attachment to a hydraulic hose; swivel of female section; and dust caps attached to the male and female sections with steel chain conforming to RR-C-271, type II, class 3, weldless, sash chain, zinc-coated. The coupling components shall be capable of withstanding a minimum of 10,000 psi internal working pressure without any evidence of leakage or failure. All coupling parts shall be of corrosion-resistant material or shall be rendered corrosion resistant by chromium or by the use of a protective coating that imparts corrosion protection to the material in the operating environment. The coupling shall be of such design that breaking the connection at the swivel will cause spring-loaded devices to seal the fluid passage in each half coupling. The spring-loaded devices shall open to permit the fluid flow when a connection is made. The exterior of the coupling shall be shaped to provide a wrench grip. The swivel shall turn freely by hand.

3.6 Set screw. Set screws shall conform to FF-S-200, type I, style 1.

3.7 Welding electrodes. Welding electrodes shall conform to AWS A5.1 or A5.18, classification and size as required.

3.8 Identification marking. The vertical connecting tool shall be metal-die stamped as shown on the drawing. Die stamping shall be legible after painting.

3.9 Treatment and painting. The vertical connecting tool, except for the couplings, screws, and threads, shall be treated and painted in accordance with MIL-T-704, type F except that the color number designation shall be 34088 or 34089 in accordance with FED-STD-595. Both the inside and the outside

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surfaces of the vertical connecting tool weldment shall be painted. Screws and threads shall be coated with preservative conforming to MIL-C-16173, grade 1.

3.10 Workmanship.

3.10.1 Steel fabrication. Steel used in the fabrication of the vertical connecting tool shall be free from laminations, kinks, and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and clipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting with a tip suitable for the thickness of metal may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. All bends shall be made with metal dies or fixtures to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. All burrs and rough edges shall be ground smooth.

3.10.2 Welders and welding.

3.10.2.1 Welders and welding operators. Before assigning any welder or welding operator to welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder or operator has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 - Structural Welding Code-Steel, ASME Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures,Welders, Brazers, and Welding and Brazing Operators.

In the event of evidence of poor welds, the Government reserves the right to require recertification of any welder or welding operator. The test results shall be made available for review by the contracting officer or the contracting officer's representative.

3.10.2.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of the ASME and AWS Codes. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire brushed clean.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise

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specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Components and material inspection. The supplier is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of preparation for delivery (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article vertical connecting tool shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The first article vertical connecting tool shall be tested as specified in 4.5.2. Failure to pass the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105 table I and table IIa. A lot shall be accepted when zero effects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

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4.5.1 Examination. The vertical connecting tools shall be examined as specified herein, for the following defects:

Major

101. Nonconformance to the dimensions shown on the drawing (see 3.1.1).
102. Components missing or not as specified (see 3.1.1).
103. Material not as specified (see 3.3).
104. Materials not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for applicable storage and operating environment (see 3.3.1).
105. Dissimilar metals as specified in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
106. Contractor does not have documentation available for identification of material, material finishes, or treatments (see 3.3.3).
107. Set screws not as specified (see 3.6).
108. Welding electrodes not as specified (see 3.7).
109. Workmanship not as specified (see 3.10).
110. Welders and welding operators not certified as specified (see 3.10.2.1).
111. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal (see 3.10.2.2).
112. Welding and welds not as specified (see 3.10.2.2).

Minor

201. Identification marking incorrect, missing, or illegible (see 3.8).
202. Treatment and painting not as specified (see 3.9).

4.5.2 Performance test. Place the vertical connecting tool in a test frame as shown in figure 1. Connect a 10,000-psi-minimum pump to the vertical connecting tool incorporating the coupling set. Install a 10,000-psi-minimum-capacity gauge in the hydraulic line between the pump and tool. Pump hydraulic fluid into the tool until the gauge in the line indicates the pressure required to produce 7 tons. Maintain this pressure on the gauge for 5 minutes. Then release the pressure and disconnect the couplings at the swivel. Determine whether there is leakage through or around the seal. Then connect the couplings again. Evidence of any of the following shall constitute failure of this test:

a. Failure of or damage to the vertical connecting tool.
b. Any leakage from the couplings.
c. Leakage through or around the seal when the couplings are disconnected at the swivel.
d. Any obstruction of the hydraulic fluid flow when the couplings are reconnected.
e. Any leakage of hydraulic fluid or evidence of damage to the hydraulic ram.

4.6 Inspection of packaging.

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4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sample size shall be determined by using MIL-STD-105 table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. One or more defects shall be cause for rejection.

113.	Materials and containers not as specified for level A or B.	Each incorrect material or container shall be considered one defect (see 5.1.1 and 5.1.2).
114.	Interior surfaces of the hydraulic ram not preserved as specified for level A or B (see 5.1.1 and 5.1.2).	
115.	Openings not sealed as specified for level A or B (see 5.1.1 and 5.1.2).	
116.	Contents not immobilized as specified for level A or B (see 5.1.1 and 5.1.2).	
117.	Box not waterproof sealed as specified for level A (see 5.1.1).	
118.	Quantities packed together exceed the weight or size limitations for level A or B (see 5.2.1 and 5.2.2).	
119.	Strapping not zinc-coated for level A (see 5.2.1).	
120.	Marking missing, illegible, incorrect, or incomplete for level A, B or C (see 5.3).	

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Interior surfaces of the hydraulic ram shall be coated with the hydraulic fluid recommended for operation and the excess fluid drained. Openings into the ram shall be sealed with tape conforming to PPP-T-60, type IV. The tool with couplings shall be packaged in a close fitting box conforming to PPP-B-636, W5c. The contents shall be immobilized within the box in a manner to prevent free movement or damage by the use of fiberboard inserts, pads, or cushions, fabricated from the same material as the box. The box shall be waterproof sealed with tape as specified for slotted style boxes in the appendix to the box specification.

5.1.2 Level B. Each tool with couplings shall be preserved and packaged as specified in 5.1.1 for level A except the box and the inserts, pads, or cushions used shall conform to PPP-B-636, type CF, class domestic, variety SW, grade 175, style optional. Box closure shall be in accordance with the appendix to the box specification.

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5.1.3 Level C. Each tool with couplings shall be preserved and packaged to afford protection against deterioration and damage from the supplier to the initial destination.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Tools preserved and packaged as specified in 5.1, shall be packed together in a close fitting box conforming to PPP-B-601, overseas type, style optional in quantities not to exceed the weight limitation of the box. Box closure shall be in accordance with the appendix to the box specification. Strapping shall be in accordance with ASTM D 3953, type 1 or 2, zinc-coated and ASTM D 4675.

5.2.2 Level B. Tools preserved and packaged as specified in 5.1, shall be packed together as specified in 5.2.1 except boxes shall be domestic type and strapping need not be zinc-coated. As an alternate the tools may be packed in a close fitting box conforming to PPP-B-640, class 2, style optional or PPP-B-636, V3c, V11c, V13c, or V15c, style optional. The gross weight or size of each box shall not exceed the limitations of the applicable box specification. Box closure shall be in accordance with the appendix to the box specification.

5.2.3 Level C. Tools preserved and packaged as specified in 5.1, shall be packed in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in a container complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The vertical connecting tool is to be used to vertically align the deck sections of the Class 60 Floating Bridge. This item is a component of the following Supply Catalogs (SC) SC 5420-97-CL-E30 and E45.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the Solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Time frame required for submission of first article (see 3.2).
d. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of

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one or more units. The contracting officer should include specific instructions in the acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation. Any changes or deviations of production vertical connecting tools from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the supplier of his obligation to furnish vertical connecting tools conforming to this specification.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Subject term key word listing.

Bridge
Erection
Floating

Custodians:
Army - ME

Preparing activity:
Army - ME

Review activity
DLA-CS

Project 5420-0184

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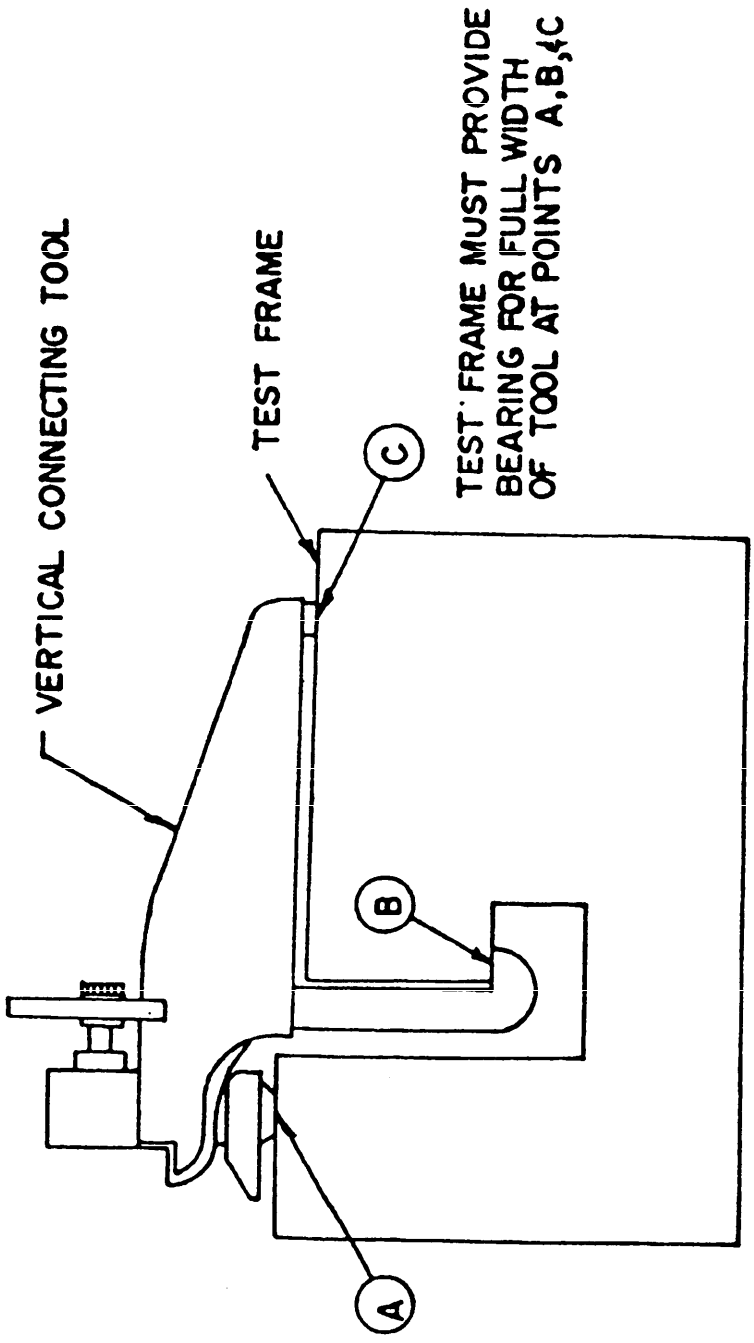


FIGURE 1. VERTICAL CONNECTING TOOL TEST FRAME.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1,2,3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4,5,6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of this letter.

NOTE: This form may not be used to request copies of documents, nor to request waives requirements in current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or do amend contractual requirements.

RECOMMENDED CHANGE:	1. DOCUMENT NUMBER MIL-T-523408	2. DOCUMENT DATE (YYMMDD) 910909

3. DOCUMENT TITLE
Tool, Vertical Connecting: Erection Equipment for Bridge, Floating, Pneumatic Float, Class 60

4. NATURE OF CHANGE (identify paragraph number and include proposed if possible. Attach extra sheets as needed).

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (LAST,FIRST MIDDLE INITIAL)	b. ORGANIZATION	
c. ADDRESS (Include zip code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. Name Gerry Harlow	b. TELEPHONE (In lude Area Code) (1) Commercial (2) AUTOVON (703) 664-2071 354-2071
ADDRESS (Include zip code) Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	If you do not receive a reply with 45 days, contact: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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